

Decoder-Recoverable Is Not Communicable: One-Clue Reference Under Constrained Vocabularies

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Abstract

Cooperative word-association games such as Codenames pose a problem of constrained reference: a speaker indicates a hidden subset of a 25-word board to a partner using one legal word and a count. We ask whether a single clue can reliably indicate an arbitrary nine-word subset, and decompose the question with a six-rung model ladder separating channel capacity, embedding geometry, vocabulary projection, and pragmatic interpretation. Under a frozen common-English vocabulary and a deterministic embedding-rank decoder, exact recovery of a nine-word subset is available for only 0.286% of board-relative target sets, by exact enumeration over 2.0×10^8 assignments, with a practical frontier near six. The limit is not geometric: synthetic vectors separate essentially every subset in 300 dimensions, so the wall is the projection from an ideal separating direction onto a legal word. In controlled pilot audits, the separations the decoder certifies do not transfer to pragmatic receivers: language-model and blinded-human guessers recover ordinary semantic clues at 23 of 24 but decoder-certified challenge clues at 0 of 44, a dissociation that survives stronger receivers and reproduces on natural game distributions. The dominant variable for natural recoverability is the availability of a clean separating concept, not the target count. A symbolic WordNet replication reproduces the collapse with no embedding. More broadly, the result bounds when embedding cosine similarity is a safe proxy for communication: least so at high cardinality and low margin, the regime where it is most tempting to trust.

1 Introduction

Cooperative word-association games such as Codenames pose a problem of *constrained reference*. A speaker observes a board of words, holds a hidden target subset in mind, and must select a single legal word that causes a receiver to identify exactly that subset. The speaker cannot point, cannot enumerate the targets, and cannot agree on an arbitrary private code: the message is one word from a shared vocabulary, and the receiver must recover the intended set from meaning. This is a compact instance of a general question about communication. How much information about a structured referent can a single symbol carry when the channel is a shared, finite, natural vocabulary?

We study the strongest form of the question that the game admits. A board holds 25 words and a hidden target subset holds up to 9. The rules invite a tempting move: find one clue that points at all nine targets at once and win on the first turn. Whether such a clue reliably exists is the question we formalize and measure. The answer turns on several constraints that are easy to

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conflate: the information capacity of a one-word channel, the geometry of the embedding space in which similarity is computed, the coverage of the legal vocabulary relative to that geometry, and the pragmatics of how a real receiver interprets a word.

The one-word channel is much weaker than the game’s nominal ceiling. Recovering an arbitrary 9-of-25 subset takes about 21 bits while one common-English word carries about 12.8, so no *universal* one-word strategy exists; the live question is per-instance existence. Under a frozen common-English vocabulary and an embedding-rank decoder, exact recovery of a 9-word subset is available for only 0.286% of board-relative target sets, by exact enumeration over 2.0×10^8 assignments, with a practical frontier near six. The shortfall is not geometric: synthetic vectors separate essentially every subset in 300 dimensions, so the wall is the *projection* from an ideal direction onto a legal word. The separations the decoder does certify then fail a second test: in a controlled audit, receivers recovered ordinary clues at 23 of 24 but decoder-certified clues at 0 of 44, so decoder-recoverability and communicability come apart. What dominates natural recoverability is the availability of a clean separating concept, not the count; both walls reproduce on natural game boards and in a symbolic model with no embedding in it.

The result matters beyond the game. Embedding cosine similarity is a common proxy for “this symbol communicates this meaning,” and the dissociation we report shows the proxy is unsafe at the high-cardinality, low-margin regime where it is most tempting to trust. The same structure recurs beyond the game, in retrieval and reranking, single-word cluster labeling, caption generation, and embedding-based evaluation; in each, optimizing the similarity score optimizes a target a human receiver does not share. The constrained-reference framing also connects the problem to referential and signaling games, where the gap between what a continuous representation permits and what a discrete shared vocabulary realizes is the object of study rather than an implementation detail.

Contributions.

1. A **model ladder** that decomposes one-clue communication into six regimes (arbitrary code-book, bounded convention, known decoder, embedding-rank decoder, synthetic vector, and public-concept / natural language), each isolating a distinct constraint, with explicit bounding relations between regimes.
2. An **exact characterization and measurement** under the known-decoder embedding-rank model: universal first-turn-9 is false (0.286% coverage), the practical exact frontier is near six, and the limit is vocabulary projection rather than embedding geometry.
3. A **dissociation result with positive controls**: clues that are certified exact recoveries under the embedding decoder do not transfer to pragmatic receivers (0 of 44), while ordinary semantic clues do (23 of 24); a stronger receiver tier replicates the zeros.
4. **Identification of the dominant variable** for natural recoverability: clean cluster availability, with target count secondary, and a threshold-like collapse at the first same-category distractor; the collapse replicates on natural random-game distributions, where cluegivers free to choose their own move retreat to counts 1 through 3.
5. A **non-embedding replication** over symbolic WordNet concepts that reproduces the same high-count collapse, defending the central claim against a purely embedding-geometric reading.

2 Related Work

2.1 Codenames as an AI Benchmark

The Codenames AI line begins with Kim et al. [11], which introduced the competition setup, agents built on WordNet and word2vec/GloVe similarity, and paired and unknown-teammate evaluation. Later agent work extended the cluegiver and guesser roles with transformer and word-association methods [10] and with language graphs and embedding scoring for multi-target clue generation [12]. A strategic-reasoning thread models the partner explicitly: recursive teammate modeling [4], adaptation under partner uncertainty [3], semantic-noise modeling [2], and Bayesian inference over partner pragmatics within a cognitive hierarchy [5]. A recent strand evaluates large language models directly, framing Codenames as an LLM benchmark [16] and as ad-hoc concept formation over target words [9]. Human and cross-cultural play is studied through Codenames Duet [14, 17], and a visual analogue appears in WinoGAViL [6].

These works build agents and measure their performance. None characterizes which target subsets are recoverable from one clue, or separates the reasons a high-count clue fails. This paper supplies that map.

2.2 Pragmatics and Decoder-Aware Speakers

Our known-decoder model, where the cluegiver knows exactly how the partner will interpret a clue, is an instance of a speaker who chooses words by modeling the listener. The Rational Speech Acts framework [8] formalizes that recursion; neural versions rerank utterances against a learned listener [1]; and signaling-game theory [13, 15] supplies the language of learned sender/receiver conventions. These describe *how* such a speaker should choose. None quantifies, for a fixed board and vocabulary, *which* target subsets it can actually reach.

2.3 Separability and Learning Theory

The synthetic-vector upper bound rests on classical separability results. Cover’s function-counting theorem [7] counts the linearly separable dichotomies of points in general position and underwrites Proposition 4; linear-separability and convex-hull characterizations give the geometric condition for a clue direction to separate targets from distractors. Several tools sharpen the failure side: VC dimension and the Sauer-Shelah lemma bound how many board subsets a vocabulary-plus-decoder concept class can realize, and separating / cover-free families and teaching dimension frame a clue vocabulary as a set system that must distinguish subsets with few messages. These results bound geometry and capacity in the abstract; they do not predict which natural clues a real vocabulary realizes, which is the projection question Section 6.3 measures.

2.4 Semantic Communication

The semantic-communication literature reframes communication around task success and meaning rather than bit-exact recovery, positioning the problem beyond Shannon channel capacity. It motivates the distinction between decoder-recoverability and communicability without supplying a board-specific feasibility result.

2.5 Gap

Existing work offers Codenames agents and benchmarks, pragmatic speaker/listener models, partner-adaptation and noisy-communication variants, and general separability and learning-theory tools.

The object none supplies is a board-relative, decoder-explicit characterization of which hidden target subsets are recoverable from one constrained semantic clue, with feasibility curves by target count and failure modes that separate information, geometry, vocabulary, and receiver limits. That is the contribution of this paper.

3 Problem Formulation

3.1 One-Clue Recoverability

Let $B = \{w_1, \dots, w_n\}$ be a board of n distinct words. A hidden target subset $T \subseteq B$ has size $|T| = k$. A cluegiver transmits a message c from a legal message set V together with a count k . A decoder D maps the board, message, count, and public history H to an ordered list of guesses:

$$D(B, c, k, H) = (g_1, g_2, \dots, g_n), \quad g_i \in B \text{ distinct.}$$

Definition 1 (one-clue recoverability). *A target subset T is one-clue recoverable under (B, V, D, H, k) if there exists a legal message $c \in V$ such that the first k guesses returned by the decoder are exactly the targets:*

$$\{g_1, \dots, g_k\} = T,$$

where $(g_1, \dots, g_k)$ is the length- k prefix of $D(B, c, k, H)$.

The first-turn instance fixes $n = 25$, the empty history $H = \emptyset$, and the target count k (with $k = 9$ the headline case).

Two distinct quantities follow from this definition, and conflating them is the source of most confusion about the game’s difficulty:

- **Per-instance existence.** For a given board and target subset (B, T) , does any legal $c \in V$ recover T ?
- **Universal guarantee.** Does a single protocol recover *every* k -subset of *every* board?

The universal guarantee is settled cheaply by counting (Section 4). The per-instance question is the one that decides whether the first-turn-9 strategy is real, and it is the question the empirical sections answer.

3.2 The Model Ladder

The recoverability of a subset depends entirely on what counts as a legal message and how the receiver decodes it. We organize the assumptions into a ladder of models, ordered from the loosest channel to the most faithful account of human play. Two rungs are idealized bounds rather than playable models, and we mark them as such. For a fixed board and target space, write R_M for the set of target subsets recoverable under model M ; the bounding relations below are stated in terms of R_M .

(1) Codebook model. The message set V is an arbitrary set of pre-agreed codewords carrying a private bijection to target subsets. Recoverability is limited only by channel size: every k -subset is recoverable once $|V| \geq \binom{n}{k}$. This rung isolates **raw channel capacity** and serves as the top bound, $R_{\text{codebook}} \supseteq R_M$ for every other M . It is a boundary case, not a semantic model.

(2) **Bounded-convention model.** A legal clue word is augmented by a pre-agreed side channel of b bits, giving capacity $|V_{\text{word}}| \cdot 2^b$. The rung isolates **how much non-semantic convention** must be added to a single word to close the gap to the codebook bound, and it interpolates between meaning-bearing clues and arbitrary codes.

(3) **Known-decoder model.** The cluegiver knows the receiver’s deterministic decoding algorithm D and may search the entire legal vocabulary. The recoverable set is exactly the image of the vocabulary under the decoder,

$$R_{\text{known}} = \{D_k(B, c, H) : c \in V\},$$

which is finite and enumerable. This rung isolates **what a vocabulary-bound but receiver-omniscient speaker can achieve**, and it is the regime relevant to AI play, where a cluegiver can simulate its partner exactly.

(4) **Embedding-rank model.** A concrete known decoder. The decoder ranks board words by cosine similarity between the clue embedding and each board-word embedding and returns the top k :

$$\text{score}(c, w) = \cos(\text{embed}(c), \text{embed}(w)),$$

$$T \text{ recoverable} \iff \min_{t \in T} \text{score}(c, t) > \max_{r \in B \setminus T} \text{score}(c, r) \quad \text{for some legal } c \in V.$$

This rung isolates the **realizable separations of a specific, widely used similarity decoder** over a legal vocabulary. It is the headline model of the paper.

(5) **Synthetic-vector model.** The message may be any vector in the embedding space, not restricted to the embedding of a legal word. Because every legal word’s embedding is itself an admissible vector, this rung upper-bounds the embedding-rank model: $R_{\text{embed}} \subseteq R_{\text{synthetic}}$. It isolates **pure geometry**, the separations the space permits independent of vocabulary. The gap $R_{\text{synthetic}} \setminus R_{\text{embed}}$ is precisely the **vocabulary-projection loss**, and measuring that gap is the mechanism claim of Section 6.

(6) **Public-concept and natural models.** The most faithful accounts of human play. In the **public-concept extension** model, the clue names a public concept (a category tag or an ontology node) and the receiver selects every board word in that concept’s board-relative extension; exact recovery requires the extension to contain no non-targets. We instantiate it two ways, with word-pack category tags and with WordNet hypernym synsets. In the **natural / pragmatic** model, the clue must work by ordinary meaning for a human or language-model receiver. The pragmatic model is the hardest to formalize; we approximate it with controlled language-model audits and reserve human panels for later validation. These rungs are not strictly nested with the embedding-rank model: a clue can be naturally communicable yet decoder-unrecoverable, and the converse, which is the dissociation the paper reports.

The ladder gives the paper its spine (Figure 1). Rungs (1) and (5) are upper bounds that establish that neither channel capacity nor geometry is the limiting factor. Rungs (3) and (4) localize the constraint to vocabulary projection. Rung (6) is the target the whole construction is meant to explain, and it is where decoder-recoverability and communicability separate.

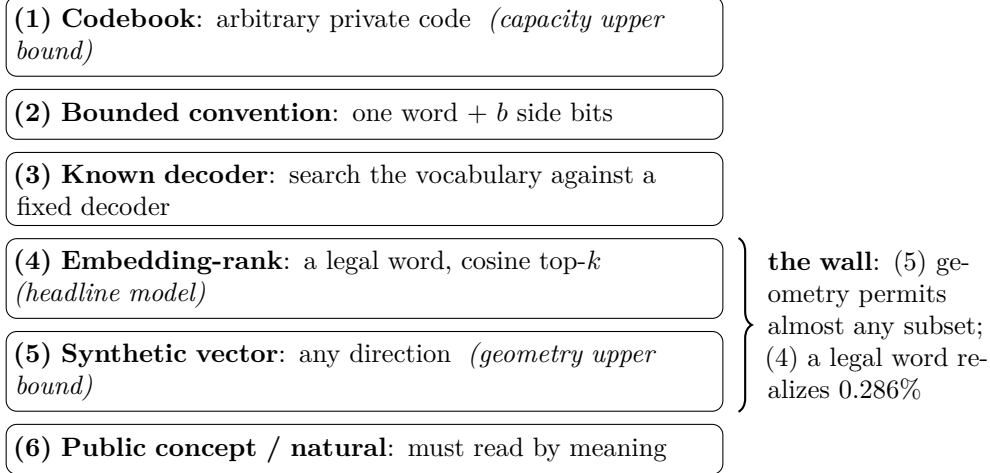


Figure 1: The model ladder, from the loosest channel (top) to the most faithful account of human play (bottom). Rungs (1) and (5) are idealized upper bounds, not playable models: neither channel capacity nor geometry forbids a first-turn-nine clue. The vocabulary-projection wall is the gap between rungs (5) and (4): a separating direction exists for almost any subset, but a legal word realizes one for only 0.286% of nine-subsets (Sections 6.1–6.3).

4 Theoretical Bounds

Four facts bound the problem before any measurement. The first settles the universal question and isolates per-instance existence as the one that remains. The second makes per-instance existence computable. The third states the exact recovery condition for the embedding-rank decoder. The fourth establishes that geometry is permissive, which localizes the limit to the vocabulary. Throughout, n is the board size and k the target count, with the headline case $n = 25$, $k = 9$.

4.1 The Information-Theoretic Floor

Proposition 1 (pigeonhole). *Any single-message protocol that guarantees exact recovery of every k -subset of an n -word board requires at least $\binom{n}{k}$ distinguishable messages.*

Proof. Distinct target subsets must map to distinct messages; if two subsets share a message, at least one is not recovered. There are $\binom{n}{k}$ subsets. $\square$

For the headline case, $\binom{25}{9} = 2,042,975$, equivalently $\log_2 \binom{25}{9} \approx 20.96$ bits. A single word drawn from the Brown-common vocabulary of 7,009 legal clues provides at most $\log_2 7009 \approx 12.78$ bits when treated as a uniform symbol channel, a deficit of about 8.19 bits. No single common-English word can serve as a universal first-turn-9 protocol: by capacity alone, one word cannot distinguish the required number of subsets.

This bound concerns the *universal* guarantee. It says nothing about whether a *given* target subset admits a recovering clue. The per-instance existence question survives Proposition 1 untouched, and it is the question the rest of the paper answers.

4.2 Decoder-Image Enumerability

Proposition 2. *For a fixed board B , finite legal vocabulary V , deterministic decoder D , history H , and count k , the recoverable target subsets are exactly the image of the vocabulary under the*

decoder,

$$R_{\text{known}} = \{D_k(B, c, H) : c \in V\},$$

and $|R_{\text{known}}| \leq \min(|V|, \binom{n}{k})$.

Proof. The map $c \mapsto D_k(B, c, H)$ is a function from V into the k -subsets of B . The recoverable subsets are its image, whose size is at most the size of the domain $|V|$ and at most the size of the codomain $\binom{n}{k}$. $\square$

Two consequences follow. First, recoverability under any known deterministic decoder is decidable by enumeration: evaluate D on every legal clue and collect the decoded subsets. This is what makes the coverage figure in Section 6.2 an exact count rather than an estimate. Second, the bound is usually loose from below, because many clues decode to the same subset, so the realized image is far smaller than $|V|$. The measured coverage quantifies how much smaller.

4.3 Rank Separability

Proposition 3. *Under the embedding-rank decoder, a clue c gives a tie-free recovery certificate for target subset T when every target outscored every non-target,*

$$m(c, T) := \min_{t \in T} \text{score}(c, t) - \max_{r \in B \setminus T} \text{score}(c, r) > 0,$$

and c certifies non-recovery when $m(c, T) < 0$. At the boundary $m(c, T) = 0$, recovery depends on the decoder's fixed tie-breaking rule. Thus $\max_{c \in V} m(c, T) > 0$ certifies recoverability over V , while $\max_{c \in V} m(c, T) < 0$ certifies non-recoverability.

Proof. The decoder returns the top k board words by score. If all k targets score strictly above all $n - k$ non-targets, the length- k prefix is exactly T . If a non-target scores strictly above at least one target, T cannot be the length- k prefix. Exact score ties at the boundary are determined by the decoder's fixed tie-breaking rule. $\square$

The quantity $\max_{c \in V} m(c, T)$ is the **projection margin** of a target subset. A positive projection margin certifies recoverability without relying on ties; a negative projection margin certifies non-recoverability because no legal clue separates T from the board. Section 6.2 reports such negative certificates directly, including a $k = 9$ subset whose best legal clue still leaves a margin of -0.069 .

4.4 The Geometry Upper Bound

Proposition 4 (synthetic separability). *If the n board vectors are in general position and $n \leq d + 1$, where d is the embedding dimension, then for every target subset T there exists a query vector v ranking all of T above all of $B \setminus T$. Equivalently, $R_{\text{synthetic}}$ is the full set of k -subsets.*

Because the decoder ranks board words by cosine, the relevant points are the *unit-normalized* board vectors: ranking by $\cos(v, e_w)$ is ranking by the dot product of v with $e_w / \|e_w\|$. The embedding packs store L2-normalized vectors (Section 5.1), so the general-position condition is stated, and checked, on exactly the vectors the decoder ranks.

Justification. Points in general position in dimension d are linearly separable under any binary labeling once $n \leq d + 1$; Cover's function-counting theorem counts the affinely separable dichotomies of n points in general position, and for $n \leq d + 1$ the count is all 2^n labelings. The separating

hyperplane’s normal induces a score direction, and the separating threshold witnesses that every target receives a higher score than every non-target. The threshold is only a proof device; it is not sent to the decoder. For $n = 25$ and $d = 300$ the condition holds with wide margin, and we confirm affine independence of the embedded board vectors on 100 of 100 sampled boards per pack (Section 6.1). $\square$

Every legal word’s embedding is itself an admissible query vector, so the synthetic model upper-bounds the embedding-rank model: $R_{\text{embed}} \subseteq R_{\text{synthetic}}$. Proposition 4 makes $R_{\text{synthetic}}$ essentially complete, so the embedding-rank shortfall measured in Section 6.2 cannot be attributed to geometry. The gap $R_{\text{synthetic}} \setminus R_{\text{embed}}$ is exactly the set of subsets whose separating direction exists but is realized by no legal word: the vocabulary-projection loss that Section 6.3 isolates.

5 Experimental Setup

5.1 Boards, Vocabulary, and Decoder

The setup has three independent pieces, each with its own pack: the *boards* (the 25 words shown), the *clue vocabulary* (the legal one-word guesses), and the *embedding* (the vectors). They are distinct; in particular the 7,009-word vocabulary below is the set of legal clues, not the boards, and the coverage figures enumerate every target subset of a board rather than sampling boards.

Boards are drawn from two packs of concrete English nouns. The primary pack is **wordnet-concrete-v0.1**; a second pack, **wordnet-concrete-band2-v0.1**, drawn from a disjoint frequency band, serves as a board-distribution sensitivity check. Each board holds 25 words, and target counts range over $k = 1..9$.

The legal clue vocabulary is **wordnet-brown-common-v0.1**, 7,009 words. It is constructed from exact WordNet lemmas filtered by a Brown-corpus frequency floor and intersected with the embedding pack, so every legal clue is a common-English word with a known vector. It is a reproducible common-English tier, not a claim about any official tabletop dictionary. Two looser tiers, a permissive lexical set and a WordNet-common set, are used only as ablations in Section 6.3.

The primary embedding is **glove-6B-300d-slim**, the 300-dimensional GloVe vectors restricted to the working vocabulary. Vectors are L2-normalized when the pack is built, so cosine similarity is computed as a dot product and every geometric statement in Section 4 refers to the unit-normalized vectors the decoder actually ranks. The decoder is the deterministic embedding-rank decoder of Section 3.2: score every board word by cosine similarity to the clue and return the top k , with a fixed tie-break. Embedding dependence is checked with a second distributional source, **glove-840B-300d-case-agg-slim**, built from Common Crawl GloVe-840B by aggregating cased variants into lowercase forms, and with the symbolic non-embedding replication in Section 6.7.

A small coverage caveat applies to the primary embedding. Two pack-1 words and six pack-2 words have no **glove-6B** vector; a board containing such a word is scored over its embedded words only, so its assignment space is $\binom{24}{k}$ or $\binom{23}{k}$ rather than $\binom{25}{k}$. Concretely, 97 of 100 pack-1 boards have all 25 words embedded and 3 have 24; for pack 2 the split is 87 / 11 / 2 boards with 25 / 24 / 23 embedded words. All denominators in Section 6 count assignments over the embedded words per board. The second embedding covers every pack word, so its boards are all 25 words; the small difference in effective board size is noted where the two embeddings are compared (Section 8).

5.2 The Pragmatic Audit Protocol

Whether a clue communicates to a real reader is not something we can compute the way we compute the embedding decoder: the natural / pragmatic model of Section 3.2 has no formula.

So we approximate a pragmatic receiver with large-language-model guessers, used as a stand-in for human receivers pending an independent human panel (Section 9), plus a blinded single-judge human author-pilot over a stratified 72-item packet (Section 6.4). A guesser sees the board, a clue, and a count, and returns an ordered list of guesses under the same recovery criterion as the decoder. Two small-tier models run each audit queue, `gpt-5.4-nano` and `claude-haiku-4-5`, with the cluegiver and guesser blinded to each other; the controls-and-challenge queue is additionally rerun with two stronger receivers, `gpt-5.4` and `claude-sonnet-4-6`, as a receiver-capability check (Section 6.4).

Every audit queue mixes two item classes. **Positive controls** are ordinary semantic clues over clean categories (for example `animal 3`, `vehicle 4`); they verify that the protocol recovers clues a competent receiver should get. **Challenge items** come from the embedding-rank decoder’s frontier and counterexample searches, and they split by certification status: 44 items carry a positive projection margin, meaning the decoder provably recovers the declared subset from that clue, and 20 items carry a negative margin, the best available legal clue for a subset the decoder cannot recover. Only the 44 certified items bear on the dissociation claim; the 20 negative-margin items are reported separately as near-threshold contrasts. Frontier items declare the count of the decoder-certified subset: for 20 of the 40 frontier items that subset is a proper contained subset (counts 4 through 7) of a 9-target key, and exact recovery is scored against the certified subset, which is what the clue provably indicates under the decoder. The controls are the load-bearing methodological check. If the guessers failed them, the audit prompt would be suspect; they do not. The guesser prompt contains only the board words, the clue, and the count; targets are never included.

Two bridge designs ground the audits in the game’s natural distribution (Section 6.6). The *stratified random-game queue* samples real first-turn boards and hidden keys under the standard ruleset and draws one target subset per count 4–9 per board from the actual key. The *best-move queue* shows the cluegiver the full 9-target key and lets it choose its own clue, count, and intended subset; generated moves are legality-filtered (single non-board clue word, chosen subset within the key, count matching the subset) before blinded audit, and exact recovery is scored against the cluegiver’s own declared subset.

Two metrics are reported. **Exact recovery** is the fraction of items where the first k guesses equal the target set. **Target-hit rate** is the fraction of all guesses that fall on targets, a partial-credit measure that separates “wrong set” from “no idea.”

5.3 Statistics

Each result is labeled by how it was computed, and the paper uses three computation classes.

Exact counts. The headline coverage figure (Section 6.2) is exact, not sampled: by Proposition 2 the recoverable set is the decoder image over the 7,009-word vocabulary, so coverage is an exact count over all board-relative target assignments. The symbolic public-concept rates of Section 6.7 are also exact, computed by inclusion-exclusion over board-relative concept extensions.

Sampled estimates. The first-turn frontier tables (Sections 6.2 and 8) sample 1,000 random 9-target keys per board (100,000 per pack), and the vocabulary-tier projection table (Section 6.3) samples 1,000 subsets per count per board. These are estimates with sampling error of order ± 0.003 on rates near 0.5 and ± 0.0002 on rates near 0.003 (one standard error). In particular, the frontier’s sampled $P(= 9)$ and the exact coverage rate estimate the same quantity under uniform random keys; any small disagreement between them is sampling error, not a model difference.

Pilot counts. The language-model audits are pilot-scale; every audit table reports raw counts and is labeled as a pilot where the per-cell sample is small. We do not attach confidence intervals to the audit rates; the 0-of-44 and 23-of-24 contrasts are reported as observed counts.

6 Results

The results run as one argument. The upper-bound rungs of the ladder are permissive, so neither channel capacity nor geometry is the constraint (6.1). Under a realistic decoder, the universal first-turn-9 strategy collapses (6.2). The collapse is a vocabulary effect, not a geometry effect (6.3). The separations the decoder does certify fail to reach pragmatic receivers (6.4). What dominates pragmatic recoverability is clean concept separation, with target count secondary (6.5). The collapse reproduces on natural random-game distributions, where cluegivers free to choose their own move retreat below the frontier (6.6). A symbolic, non-embedding model reproduces the same shape (6.7).

6.1 The Upper Bounds Are Permissive

Neither channel capacity nor geometry forbids first-turn-9. The codebook rung makes every 9-subset recoverable once the channel carries the 2,042,975 messages of Proposition 1, so the problem is not impossible in principle. The synthetic-vector rung makes it geometrically available: across 100 sampled boards per pack, the embedded board vectors (25 per board on most boards; 24 on three pack-1 boards and 23–24 on thirteen pack-2 boards, per the coverage caveat of Section 5.1) were affinely independent in 100 of 100 cases for each pack, so by Proposition 4 a separating query direction exists for essentially every target subset. The interesting failure therefore lives strictly between “any vector” and “a legal word.”

Caveat. General position is a property of the sampled GloVe boards; a degenerate embedding could in principle violate it, which is why the condition is checked rather than assumed.

6.2 Universal First-Turn-9 Collapses Under a Realistic Decoder

Under a frozen common vocabulary and the embedding-rank decoder, almost no 9-subset is recoverable. Exact enumeration over the decoder image gives a $k = 9$ coverage of $578,424/202,091,087 = 0.286\%$ on the primary pack: fewer than three in a thousand board-relative 9-subsets admit any legal recovering clue. The denominator counts 9-subsets of the embedded words per board: 97 boards contribute $\binom{25}{9}$ and three contribute $\binom{24}{9}$ (Section 5.1). A direct certificate makes the failure concrete. On a 25-word board with targets {barn, builder, call, colonel, educator, pit, product, specialist, wood}, the best legal clue (darling) still leaves a projection margin of -0.069 , so by Proposition 3 no legal clue separates the set. The certificate is a statement about the explicit board word list recorded in the artifact (the board was drawn from an earlier revision of the word pack); it was rechecked against the current legal vocabulary, 6,869 legal clues for that board, with the same best clue and margin.

The practical frontier sits near six. Given a random 9-target key (1,000 sampled keys per board, 100,000 per pack; sampled estimate, Section 5.3), the largest exactly recoverable contained subset has the following distribution:

pack	$E[\text{best exact count}]$	$P(\geq 6)$	$P(\geq 7)$	$P(\geq 8)$	$P(= 9)$
wordnet-concrete-v0.1	5.87	0.642	0.221	0.038	0.0030
wordnet-concrete-band2-v0.1	5.91	0.666	0.235	0.041	0.0032

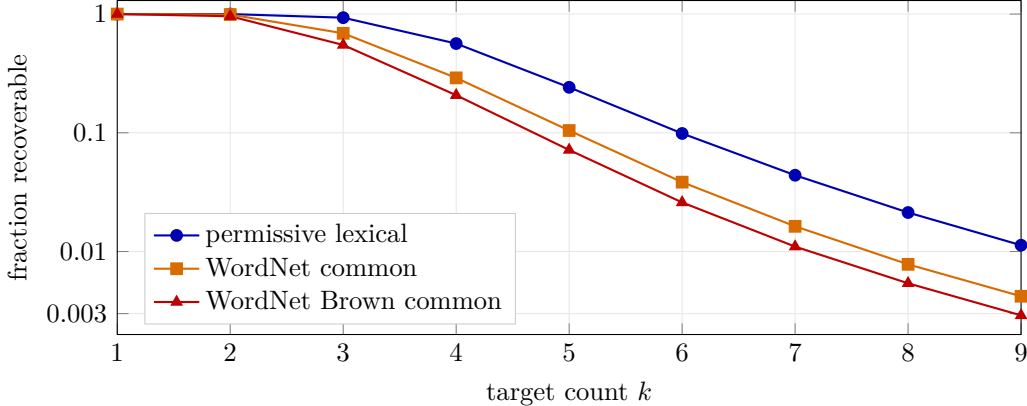


Figure 2: Coverage collapses with target count, and a stricter vocabulary lowers the whole curve. Fraction of board-relative target subsets a legal clue can separate (positive projection margin) by count k , for three clue-vocabulary tiers, primary pack, log scale. The common-English tier reaches 0.0029 at $k = 9$ (the 0.286% headline); each step toward a more natural vocabulary is the projection effect of this section.

The exact frontier is usually five or six, occasionally seven or eight, and rarely nine. The sampled $P(= 9)$ column and the exact coverage rate above estimate the same quantity; they agree within sampling error.

Caveat. The coverage figure is exact and the frontier rows are sampled estimates, in both cases for one embedding and two board packs; generalization to other embeddings is addressed in 6.7 and Section 8.

6.3 The Bottleneck Is Vocabulary Projection

The collapse is the gap between what the geometry permits and what a legal word realizes. Synthetic vectors recover essentially everything (6.1); legal words recover 0.286% at $k = 9$ (6.2). The difference, $R_{\text{synthetic}} \setminus R_{\text{embed}}$, is the set of subsets whose separating direction exists but is named by no legal clue. Tightening the vocabulary deepens the collapse monotonically; since the geometry is identical across tiers, the effect is lexical (Figure 2; sampled positive-margin rates, 1,000 subsets per count per board; Section 5.3):

clue tier	$k=3$ recoverable	$k=5$ recoverable	$k=9$ recoverable
permissive lexical	0.932	0.242	0.0113
WordNet common	0.689	0.105	0.0042
WordNet Brown common	0.549	0.072	0.0029

Each step toward a stricter, more natural vocabulary removes recoverable subsets, most sharply at high counts. Simple static remedies do not change the high-count picture: an all-but-the-top-component transform raises low- k coverage but leaves $k = 9$ near 0.003 on both packs. The wall is lexical coverage of the embedding’s separating directions, and it cannot be transformed away while the clue must be a legal word.

Caveat. The permissive-lexical tier is an optimistic ablation, not a playable clue model; it bounds the effect of vocabulary loosening rather than describing real play.

6.4 Clues Optimal Under the Decoder Do Not Reach Pragmatic Receivers

Decoder-recoverability and natural communicability come apart. In the controlled audit, language-model guessers recovered ordinary clues nearly perfectly and recovered decoder-certified challenge clues at zero. The challenge items split by certification status (Section 5.2): the 44 decoder-certified items each carry a positive projection margin, so the decoder provably recovers the declared subset from the clue, and they are the items that bear on the dissociation; the 20 negative-margin items are near-threshold contrasts:

item class	projection margin	exact recovery	target-hit rate
positive controls	—	23/24	98.9%
decoder-certified challenge	+0.0001 to +0.033	0/44	37.7%
best-available negative-margin	−0.083 to −0.001	0/20	35.6%

The split holds across both models, so it is not an artifact of one guesser:

model	controls exact	certified challenge exact	negative-margin exact
<code>gpt-5.4-nano</code>	11/12	0/22	0/10
<code>claude-haiku-4-5</code>	12/12	0/22	0/10

The controls rule out a broken protocol: the same machinery that recovers **animal 3** and **vehicle 4** recovers none of the decoder-certified challenge clues. The lone control miss is meaning-driven: under **car 4** the reader takes **truck** over the intended **driver**, a same-category near-swap that recurs at the stronger receiver tier. A clue that maximizes the projection margin is exploiting how a specific ranking algorithm orders words, not communicating the target set as meaning. The certified margins are also small in absolute terms (at most +0.033), which is consistent with the discussion’s point that the dissociation concentrates in the low-margin regime. A stronger pilot points the same way: when language-model cluegivers were asked to generate their own natural single-word clues for $k = 9$ challenge targets, blinded guessers recovered 0 of 40 exactly, even as the clues read as more natural.

A receiver-tier check bounds the capability confound. If the challenge failures reflected weak receivers rather than non-communicability, stronger receivers should recover some certified clues. Rerunning the identical 44-item queue with substantially stronger guessers reproduces the zeros exactly:

model	controls exact	certified challenge exact	negative-margin exact
<code>gpt-5.4</code>	11/12	0/22	0/10
<code>claude-sonnet-4-6</code>	12/12	0/22	0/10

Partial signal moves slightly (certified target-hit rate 42.0% at the stronger tier versus 37.7% at the small tier), but no decoder-certified clue becomes exactly recoverable when receiver capability scales. The dissociation does not close with receiver strength.

A blinded human author-pilot reproduces the pattern. The 72-item publication-transfer packet (12 items per stratum, targets and strata hidden from the labeler) was labeled by the first author as a blinded guesser. Decoder-certified challenge clues recovered 0 of 12, with the first guess already a non-target in 6 of the 12 items; LLM-generated and arbitrary-subset challenge clues recovered 0 of 24. The cluster strata, which evidence the labeler’s competence, land almost

exactly on the LLM rates: clean clusters 6/12 exact with an 85.1% target-hit rate (LLM pilots: 50% and 85.9%), and same-tag interference 0/12 with 68.4% hits (LLM: 0% and 68.3%). One hypothesis-aware author is not an independent panel, so under the study’s own wording gate this remains pilot evidence; it does, however, weigh against the reading that the dissociation is a quirk of LLM receivers that human receivers would not share.

Caveat. The audit is pilot-scale ($N = 24$ controls and $N = 64$ challenge per receiver tier, two tiers), and the pragmatic receivers are language models standing in for humans; a human panel is future work (Section 9). The contrast is large enough to report as observed counts, not as a powered rate.

6.5 The Dominant Variable Is Cluster Availability; Count Is Secondary

What decides natural recoverability is primarily whether a clean separating concept exists; target count matters, but only secondarily once such a concept is available. Lowering the count on hard arbitrary subsets does not rescue them: across counts 4 through 9 on the challenge distribution, generated natural clues recovered 0 of 96 exactly. Yet explicit semantic clusters transfer at far higher rates, and a small number of same-category distractors destroys them:

target distribution	exact recovery	target-hit rate
hard arbitrary subsets	0/96	37.2%
explicit semantic clusters	24/48	85.9%
clusters + 2 same-tag distractors	0/48	68.3%

Concretely, a clean separating concept is a sharp category that names the targets and nothing else on the board: four animals, with no other animal among the 25 words, clued by **animal**. This is stronger than geometric separability: the synthetic upper bound (Section 6.1) shows a separating *direction* exists for almost any subset, but a clean concept also needs a single legal *word* to realize that direction and a real reader to read it. The per-category breakdown later in this section shows the variable directly.

Within the clean clusters, count retains a secondary effect: exact recovery is 16/24 at counts 4 through 6 and 8/24 at counts 7 through 9, including 2/8 at count nine. Cluster availability moves arbitrary high-count subsets from 0% to roughly a third recovered; count then roughly halves the rate within clean clusters. The per-category pattern points the same way: specific categories with sharp board-relative extensions transfer well (**vehicle** 6/6, **clothing** 5/6, **weapon** 4/4), while broad categories transfer at zero (**place**, **structure**, **substance** 0/8 combined), which is the clean-separating-concept variable showing through the category inventory itself.

The collapse is threshold-like, and the first distractor does most of the damage. Holding the target families fixed and adding same-tag non-targets to the board:

same-tag distractors	exact recovery
0	15/48
1	3/48
2	2/48
3	3/48
4	1/48

The sweep’s zero-distractor baseline (15/48) is lower than the cluster table above (24/48) because the two designs draw different category mixes: the sweep requires families large enough to

supply four added same-tag distractors, which biases it toward broad categories, exactly the ones that transfer worst. On the categories the designs share, the rates agree (**animal** 3/6 in both, **container** 0/6 in both).

Exact recovery falls off the cliff at the first distractor while partial signal (target-hit rate) degrades gradually, which marks a distinction between clean category membership and exact category separation on the whole board. Letting a target-aware cluegiver propose narrower clues than the broad category label recovered a few more cases at counts 4 through 6 (7/48) but left counts 7 through 9 at 0 of 48, so a better contextual cluegiver does not work around same-category interference inside the single-word model.

The blinded human author-pilot of Section 6.4 reproduces both cluster strata nearly exactly (clean clusters 6/12 with 85.1% hits; same-tag interference 0/12 with 68.4% hits), which makes the cluster-availability and interference claims less dependent on the LLM proxy, within the author-pilot caveat stated there.

Caveat. The cluster and distractor sweeps are constructed existence/interference designs at audit scale ($N = 48$ to 96 per cell), reported as counts.

6.6 The Collapse Reproduces on Natural Game Distributions

Constructed challenge sets are not doing the work: the same high-count collapse appears on real game boards, and cluegivers free to choose their own move retreat below the frontier. All preceding audits use constructed target distributions. Two bridge pilots ground the result in the game’s natural state distribution.

In the **stratified random-game bridge**, eight first-turn boards per pack were generated under the standard hidden-key ruleset, with one target subset sampled from the first player’s actual key at each count 4 through 9 per board. Target-aware LLM cluegivers wrote a natural clue for each subset, and crossed blinded guessers audited every clue (two generators by two auditors). Exact recovery was zero everywhere:

pack	audit rows	exact	target-hit rate
wordnet-concrete-v0.1 , 8 boards	192	0/192	38.9%
wordnet-concrete-band2-v0.1 , 8 boards	192	0/192	33.9%
combined	384	0/384	36.4%

Recovery was 0 of 64 at every count from 4 through 9, and the failures are immediate rather than near misses: 289 of 384 audits (75%) placed a non-target at guess rank 1 or 2. On natural boards, as on constructed ones, partial semantic signal survives while exact recovery does not.

In the **best-move pilot**, the cluegiver instead sees the full 9-target key and chooses its own clue, count, and intended subset, mirroring a real first move. Given the choice, the tested cluegivers retreat to low counts: across 32 boards (16 per pack), 50 of 64 generated moves were legal, with chosen counts concentrated at 1 through 3. Over the 100 valid crossed-audit rows, exact recovery was 28/100, all of it at counts 1 through 3 (11/20 at count 1, 9/24 at count 2, 8/28 at count 3); every audit at count 4 or higher failed (0/28), including all four count-9 audits. The cluegivers’ own revealed preference lands at the same frontier the decoder-image analysis locates near six and pragmatic transfer locates lower still.

A vocabulary-cleanliness check argues against noisy board words as the explanation: rerunning the best-move bridge on a 300-word curated common-noun pack (**playable-common-v0.1**) kept the same shape, 6/36 valid-audit exact, all at counts 1 through 3, and 0/16 at counts 4 and above.

Caveat. Both bridges are pilot-scale and use small-tier LLM cluegivers and guessers; counts are reported as observed. They measure the tested cluegiver/guesser pairs on the natural distribution, not a bound over all possible natural play.

6.7 Symbolic Public Concepts Reproduce the Shape

The high-count collapse also appears in a non-embedding public-concept model. Replacing the embedding decoder with a symbolic public-concept decoder, where a clue is a public category or ontology node and the receiver selects exactly the board words in that concept’s extension, gives the same picture from a non-embedding direction. Computed exactly over all $\binom{25}{9}$ first-turn assignments per board:

concept model	$E[\text{best exact count}]$	exact $P(= 9)$ per assignment
word-pack tags	≈ 1.0	0 (exact)
WordNet hypernym, 1 sense/word	1.39	$\approx 1.1 \times 10^{-7}$
WordNet hypernym, 3 senses/word	1.72	$\approx 1.0 \times 10^{-7}$

Values shown are for the primary pack; the second pack agrees (one-sense $E[\text{best}]$ 1.39, exact $P(= 9) \approx 1.2 \times 10^{-7}$; three-sense 1.62 and $\approx 1.4 \times 10^{-7}$).

A richer public ontology raises the low-count baseline but leaves exact first-turn-9 at roughly one in ten million per random assignment. The rare size-9 concept extensions that do exist are qualitatively broad: across saved examples (20 of the 21–28 extensions per run are archived), every exact size-9 extension was labeled **artifact** or **organism**, so the rare successes are board-count coincidences for very general classes, not narrow high-count concepts. A symbolic model with no embedding anywhere in it therefore reproduces the high-count collapse, though it does not by itself test another distributional embedding.

Caveat. The public-concept decoder is stricter than human pragmatics (it selects every matching board word), so it under-counts what a flexible human clue could do; it is a clean lower bound on concept-based recoverability, not a model of natural play.

7 Discussion

The results compose into a layered account. Channel capacity and geometry are both permissive (Section 6.1): the codebook bound is finite, and in 300 dimensions a separating direction exists for essentially every target subset. The first wall is vocabulary projection: the legal words realize only a sparse subset of those directions, and tightening the vocabulary toward natural words widens the gap (Sections 6.2, 6.3). A second wall stands behind it. Even the separations a legal clue realizes are often artifacts a margin-maximizer exploits rather than meanings, and they do not survive a pragmatic receiver (Section 6.4); what does survive is a clean separating concept on the whole board, which a single same-category distractor destroys (Section 6.5). Neither wall is an artifact of constructed challenge sets: both hold on natural game boards (Section 6.6) and in a symbolic model with no embedding in it (Section 6.7).

Two implications follow for practice. First, embedding cosine similarity is an unsafe proxy for “this symbol communicates this set,” and the failure concentrates where it would do the most damage, at high cardinality and low margin, the regime where the proxy still reads as confident. The same structure recurs wherever a single embedding-decoded signal must pick out an exact set: retrieval and reranking, single-word cluster or category labeling, referring-expression and caption

generation, and embedding-based evaluation. In each, optimizing the similarity score optimizes a target the human receiver does not share, and the dissociation we measure here predicts the same gap. Second, the variable worth controlling is concept separation before count: count is a much weaker predictor of difficulty once a clean separating concept is in view, whether the task is clue generation, label generation, or benchmark design.

Read as a statement about constrained reference, the paper isolates a general bottleneck. When a speaker must indicate a structured referent through a single symbol drawn from a shared, finite vocabulary, the limit is the projection from an ideal meaning onto a legal, board-separating word. The geometry of the underlying representation can be permissive while the vocabulary that names it is not.

8 Limitations

The headline measurements are conditioned on a frozen decoder, legal vocabulary, and board-generation process. Three things bound embedding and board dependence. First, the two board packs are drawn from disjoint frequency bands and agree throughout. Second, a second distributional embedding source, GloVe-840B Common Crawl with cased variants aggregated to lowercase forms, reproduces the same high-count collapse:

embedding	pack	exact $k=9$ rec.	$E[\text{best}]$	$P(\geq 6)$	$P(\geq 8)$	$P(= 9)$
GloVe-6B	pack 1	0.00286	5.87	0.642	0.0379	0.00303
GloVe-6B	pack 2	0.00303	5.91	0.666	0.0410	0.00319
GloVe-840B case-agg.	pack 1	0.00300	5.98	0.706	0.0409	0.00284
GloVe-840B case-agg.	pack 2	0.00300	6.01	0.723	0.0419	0.00309

The “exact $k=9$ recoverable” column is an exact count; the frontier columns are sampled estimates (Section 5.3). One comparability caveat: the 840B pack embeds every board word, while a few GloVe-6B boards run at 24 or 23 embedded words (Section 5.1), so the two embeddings are measured on slightly different effective boards. This does not make the result universal over all embedding algorithms, but it does rule out a simple “the 6B vector source is uniquely bad” explanation. Third, the symbolic public-concept replication (6.7) reaches the same high-count collapse with no embedding involved, so a purely GloVe-geometric explanation is incomplete.

The pragmatic receiver is primarily a language model standing in for a human. The controls (6.4) establish that the audit recovers clues a competent receiver should get, and two receiver-validity checks bound the proxy risk: receivers at two capability tiers, **gpt-5.4-nano/claude-haiku-4-5** and **gpt-5.4/claude-sonnet-4-6**, produce the same zero exact transfer on decoder-certified items, and a blinded human author-pilot over the stratified 72-item packet reproduces every stratum’s pattern, including the cluster and interference rates (6.4). The human evidence is a single hypothesis-aware author, so all human-transfer statements remain pilot-grade; an independent human panel is the proper test of natural communicability and is future work (Section 9). The audit cells are pilot-scale; the large contrasts (23 of 24 versus 0 of 44 decoder-certified per tier, the 0-of-384 random-game bridge, and the cluster and distractor sweeps) are reported as observed counts rather than powered rates, and the small-count rows should be read as directional. The work is English-only and uses a single decoder algorithm; the embedding-rank top- k decoder is one natural choice, and other deterministic decoders would induce different images, though the projection argument applies to any vocabulary-constrained decoder. The public-concept decoder of 6.7 is stricter than human pragmatics, since it selects every board word in a concept’s extension; it is a lower bound on concept-based recoverability, not a model of flexible natural clueing.

9 Conclusion and Future Work

A single word from a shared, finite vocabulary cannot reliably indicate an arbitrary nine-of-twenty-five target subset, and the reason is specific. Capacity and geometry are permissive; the limit is the projection from an ideal separating direction onto a legal word, and the separations a decoder does certify often fail to reach a pragmatic receiver. The variable that dominates natural recoverability is clean concept separation on the board, with target count secondary. These statements are exact under the tested known decoder, reproduced symbolically in a stricter public-concept model, and stable across two board packs and a second distributional embedding source.

Four directions follow. An independent human panel would replace the language-model receiver in the transfer and cluster audits; a blinded single-author pilot already reproduces the dissociation and the cluster threshold (6.4), and a panel of independent labelers on the same packet is the remaining step to a non-pilot human claim. A formal natural-clue model, most plausibly in the Rational Speech Acts family, would give the pragmatic rung of the ladder a closed form rather than an audit proxy. Distributional bounds in terms of board size, vocabulary size, embedding dimension, and semantic density would turn the measured curves into predictions. Extending the model from one-clue recoverability to full-game value is a separate problem for which this map is a prerequisite. Because the reliable one-clue frontier sits near six rather than nine, optimal play is inherently multi-turn and strategic: a cluegiver must choose which few targets to indicate each turn, trading coverage against the risk that a clue broad enough to reach its targets also ranks an opponent’s word, a bystander, or the assassin highly. The projection margin already supplies the language for that trade-off, the margin over one’s own targets against proximity to dangerous words, and competitive game-value analysis on top of the recoverability map is a natural next step that we leave to future work.

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